

THE ELEMENTS

Shiny gold, tough iron, smelly chlorine, and invisible oxygen—what do they have in common? They are all elements: substances made of only one type of atom that cannot be broken down into a simpler substance. But they can combine with other elements to form new substances, known as compounds. Everything around us is made up of elements, either in pure form or combined. Water, for example, is made of the elements hydrogen and oxygen. There are 118 known elements, of which around 90 exist naturally. The rest have been created in laboratory experiments.

In 1869, the Russian scientist Dmitri Mendeleev came up with a system for how to sort and classify all the elements. In his chart, the atomic number increases left to right, starting at the top left with hydrogen, with an atomic number of 1. Arranging elements in rows and columns reveals patterns. For example, elements from the same column, or group, react in similar ways and form a part of similar compounds.

This is the number of protons in the atom's nucleus. The element iron has an atomic number of 26, which means it has 26 protons (and 26 electrons).

An atom's mass is how many protons and neutrons it has. This number shows the relative atomic mass (the average mass of all an element's atoms, see p.13).

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An element has the same symbol all over the world, while the name can be different in different languages.

An element's place in the table is decided by its atomic number. Each element has a "tile" showing its atomic number, its chemical symbol, and its atomic weight (weights in brackets are estimates for unstable elements). The symbol is an abbreviation of the element's original name. This name was often invented by the person who discovered the element.

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The periodic table

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Elemental information

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Lanthanides and actinides

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